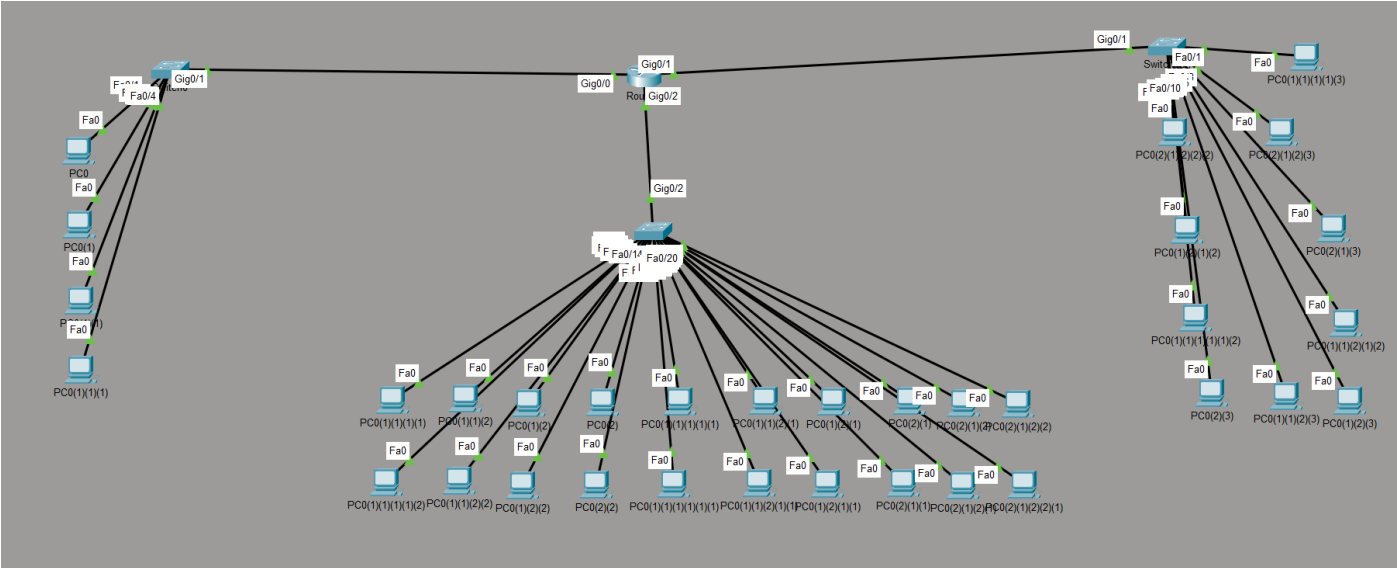




VLSM

IPv4

LAN 1

20 PCs + Router + switch = 24 hosts

Networks: 192.168.10.0/27

Mask: 255.255.224

F. Usable: 192.168.10.1

L. Usable: 192.168.10.30

Broadcast: 192.168.10.31

Gateway: 192.168.10.1

LAN 3

10 PCs + Router + switch = 14 hosts

Networks: 192.168.10.32/28

Mask: 255.255.240

F. Usable: 192.168.10.33

L. Usable: 192.168.10.46

Broadcast: 192.168.10.47

Gateway: 192.168.10.33

LAN 2

4 PCs + Router + switch = 6 hosts

Network: 192.168.10.48/29

Mask: 255.255.248

F. Usable: 192.168.10.49

L. Usable: 192.168.10.54

Broadcast: 192.168.10.63

Gateway: 192.168.10.49

IPv6

LAN 1

2001::DB8:ACAD:1::/64

Gateway: 2001::DB8:ACAD:1::1

LAN 2

2001::DB8:ACAD:2::/64

Gateway: 2001::DB8:ACAD:2::1

LAN 3

2001::DB8:ACAD:3::/64

Gateway: 2001::DB8:ACAD:3::1

```

R1#
R1#show ipv6 interface brief
GigabitEthernet0/0      [up/up]
    FE80::202:17FF:FE5B:3501
    2001:DB8:ACAD:1::1
GigabitEthernet0/1      [up/up]
    FE80::202:17FF:FE5B:3502
    2001:DB8:ACAD:3::1
GigabitEthernet0/2      [up/up]
    FE80::202:17FF:FE5B:3503
    2001:DB8:ACAD:2::1
Vlan1                    [administratively down/down]
    unassigned
R1#

```

```

R1#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0    192.168.10.49   YES manual up          up
GigabitEthernet0/1    192.168.10.33   YES manual up          up
GigabitEthernet0/2    192.168.10.1    YES manual up          up
Vlan1              unassigned      YES unset  administratively down down
R1#

```

```

R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

```

Gateway of last resort is not set

```

    192.168.10.0/24 is variably subnetted, 6 subnets, 4 masks
C       192.168.10.0/27 is directly connected, GigabitEthernet0/2
L       192.168.10.1/32 is directly connected, GigabitEthernet0/2
C       192.168.10.32/28 is directly connected, GigabitEthernet0/1
L       192.168.10.33/32 is directly connected, GigabitEthernet0/1
C       192.168.10.48/29 is directly connected, GigabitEthernet0/0
L       192.168.10.49/32 is directly connected, GigabitEthernet0/0

```

```

R1#

```